Unit: Unit 6: Systems of Linear Equations and Inequalities **Name:** ____________________
Topic: Applications of Systems of Inequalities **Date:** ____________

Quiz (Carolina Reaper)

Open Ended Questions (FRQ)

Q9. A bike manufacturer produces two models, the Speedster and the Cruiser. The Speedster requires 2 hours of assembly and 3 hours of painting, while the Cruiser requires 3 hours of assembly and 2 hours of painting. The total available assembly time is 240 hours and the total available painting time is 300 hours. Let x be the number of Speedsters and y be the number of Cruisers. Write a system of inequalities representing the constraints.

Q10. A farmer has 300 acres of land to plant with wheat and corn. Each acre of wheat requires $100 of capital and each acre of corn requires 120$ of capital. The farmer has $36,000 of capital available. Let x be the number of acres of wheat and y be the number of acres of corn. Find the maxi$ and the profit per acre of corn is 250.

Chef's Tips

- Tip 1: Check for corner points in the feasible region
- Tip 2: Verify that the solution satisfies all constraints
- Tip 3: Graph the system of inequalities to visualize the feasible region